Carbon Credit Auction Clearing via Ising Optimisation

Quantum Dice Trinity Challenge 2026 Stage 2 Submission by Ballu Di Team

1. Problem and Data

Problem Context

Under the EU Emissions Trading System (EU ETS), the European Commission distributes a fixed system-wide cap of carbon allowances across heterogeneous credit categories: general allowances (EUA) for stationary and maritime installations, aviation allowances (EUAA), and member-state sub-quotas for 25 participating states. Each category carries distinct eligibility rules and sector-specific supply constraints. From 2025, EUAA volumes are integrated into EUA spot auctions, creating explicit cross-type allocation decisions.

The clearing problem is combinatorial: N bids compete for volume across M credit types differentiated by sector eligibility and supply cap. This is weighted k-set packing, for which Hazan, Safra and Schwartz [1] proved no polynomial-time approximation within $o(k/\log k)$ unless P = NP, motivating a probabilistic computing approach.

Data Source

Auction session statistics are taken from the EEX Primary Auction Spot Report 2025, publicly available at eex.com. The dataset covers 213 sessions across four credit types from January to December 2025, providing per-session clearing price, supply volume, number of bids submitted, average bid size, standard deviation of bid volume, and cover ratio.

Data Limitations and Bid Reconstruction

Individual bid data is confidential under EU ETS regulations. We reconstruct a synthetic bid population whose statistical properties match the published session aggregates exactly, standard practice in computational economics when microdata is unavailable [6].

For each session, n_{bids} synthetic bids are generated. Bid quantity q_i is drawn from a Pareto distribution parameterised by the published mean μ_q and standard deviation σ_q:

$$\alpha = 2 + \frac{\mu_q^2}{\sigma_q^2 - \mu_q^2}, \qquad x_m = \frac{\mu_q(\alpha - 1)}{\alpha} \tag{1}$$

Log-normal was considered but rejected: EU ETS bid volumes exhibit coefficient of variation $\sigma_q/\mu_q > 3$, a regime where log-normal severely underestimates the heavy tail. Pareto is the standard distribution for heavy-tailed positive auction data.

Bid price p_i is drawn from a triangular distribution with support $[p_{\min}, p_{\max}]$ and mode at the published clearing price p^*:

$$p_i \sim \text{Triangular}(p_{\min},\, p^*,\, p_{\max}) \tag{2}$$

Bid value is $v_i = q_i p_i$. After reconstruction, quantities are rescaled so total bid volume matches the real session cover ratio $\rho = \sum_i q_i / S_\tau$, ensuring the supply constraint is maintained. Observed cover ratios range from 1.54 (EUA) to 2.06 (EUA_DE), confirming demand substantially exceeds supply.

A single representative session per credit type is selected as the median session by date, yielding 321 reconstructed bids across four types. Package bid linking is described in Section 2.

2. Formulation

We retain the binary variable $x_i \in \{0, 1\}$ per bid and welfare objective $\max \sum_i v_i x_i$ from Stage 1. Two structural changes are made to the Hamiltonian.

Removal of H_{excl}

Stage 1 included a pairwise exclusivity penalty $H_{\text{excl}} = B \sum_j \sum_{i<k} a_{ij} a_{kj} x_i x_k$ to prevent two bids claiming the same physical lot j. Under EU ETS, however, the commodity being auctioned is fungible tCO_2 volume drawn from a common pool there is no unique lot j exclusive to one bidder. Any number of bids may be accepted provided total accepted volume does not exceed supply S_τ. This is enforced entirely by H_{cap}, rendering H_{excl} redundant. In practice, retaining it imposed 4,950 spurious antiferromagnetic couplings across same-type bids, preventing SA from accepting more than two bids regardless of supply headroom.

Introduction of H_{pkg}

Aviation operators holding EUAA allowances also participate in EUA sessions to cover non-aviation installations. Such a bidder values an EUA–EUAA bundle jointly more than the sum of its parts: obtaining only one type leaves a compliance gap. This complementary valuation introduces a synergy premium $w_{ik} > 0$ when bids i (EUA) and k (EUAA) are accepted together, with $w_{ik} = 0.15 \min(v_i, v_k)$. We model 30% of EUAA bidders as holding a linked EUA bid, giving 14 complementary pairs in the toy instance. The bonus term is:

$$H_{\text{pkg}} = -A \sum_{(i,k) \in \mathcal{P}} w_{ik}\, x_i x_k \tag{3}$$

This adds off-diagonal attractive couplings $J_{ik} = -A w_{ik}$, making bid acceptance decisions non-separable: the optimal choice for bid i depends on whether its partner k is accepted. The ground state of H therefore reflects joint compliance structure across credit types rather than independent per-type welfare maximisation.

Updated Hamiltonian

The full Hamiltonian is $H = H_{\text{obj}} + H_{\text{pkg}} + H_{\text{cap}}$:

$$H_{\text{obj}} = -A \sum_i v_i x_i \tag{4}$$

$$H_{\text{pkg}} = -A \sum_{(i,k) \in \mathcal{P}} w_{ik}\, x_i x_k \tag{5}$$

$$H_{\text{cap}} = C \sum_\tau \left(\sum_{\{i:\tau(i)=\tau\}} q_i x_i - S_\tau \right)^2 \tag{6}$$

Penalty Coefficient Derivation

Stage 1 stated the dominance condition $C > A \cdot v_{\max}$, which assumes each constraint coefficient q_i is $O(1)$. This fails when individual bids are small fractions of supply in EU ETS, $q_i \approx S_\tau/70$ after normalization.

The correct bound follows from the SA spin-flip condition. When SA considers accepting bid i given n_{acc} bids already accepted of the same type, the energy change is:

$$\Delta E = Q_{ii} + 2 \sum_{k \text{ acc}} Q_{ik} \tag{7}$$

For the flip to be rejected ($\Delta E > 0$), accumulated off-diagonal pushback from H_{cap} must exceed the combined reward from H_{obj} and any package coupling from H_{pkg}:

$$2C\, q_i \sum_{k \text{ acc}} q_k \;>\; A\, v_i + A\, w_{ik}\, \mathbf{1}[k \in \mathcal{P}_i] \tag{8}$$

where $\mathbf{1}[k \in \mathcal{P}_i]$ indicates bid i has an accepted package partner. Taking the worst case $v_i = v_{\max}$, $w_{ik} = w_{\max}$, $\sum_k q_k \approx n_{\text{acc}}\, \bar{q}$, and normalising by S_τ so $S_\tau \to 1$:

$$C \;>\; \frac{A\,(v_{\max} + w_{\max})}{2\, q_i\, n_{\text{acc}}\, \bar{q}} \tag{9}$$

With $n_{\text{acc}} \approx 161$, $\bar{q} \approx 0.022$, $q_i \approx \bar{q}$, $A = v_{\max} = 1$, and $w_{\max} \approx 0.015$ after normalisation, equation (9) gives $C > (1 + 0.015)/(2 \times 0.022 \times 161 \times 0.022) \approx 6.5$. The bound is a necessary condition derived at the mean: bids drawn from a heavy-tailed Pareto distribution can be significantly smaller than $\bar{q}$, reducing the effective off-diagonal pushback for those bids and requiring a larger margin. We therefore set $C = 100(v_{\max} + w_{\max} + 1) \approx 200$, a $30\times$ safety margin above the mean-field bound, sufficient to enforce feasibility across the full bid size distribution.

A 2% supply buffer ($S_\tau \to 0.98\, S_\tau$ in the QUBO) absorbs discretisation error from the heavy-tailed bid size distribution, costing less than 0.5% in welfare.

3. Results

Solvers

Simulated annealing is implemented via `dwave-neal` [7] with 200 independent reads and 1,000 sweeps per read. The classical baseline uses integer programming (IP) via PuLP/CBC, which solves the problem exactly and instantaneously at this scale. All quantities are normalised per credit type before building the QUBO; welfare is reported in real euros from raw bid values.

Final Results

Figure 1 shows SA welfare tracking IP welfare closely across sessions, with a mean gap of approximately 2%

4. Limitations and Extensions

QUBO Scaling

The squared penalty $C(\sum q_i x_i - S_\tau)^2$ is structurally problematic when $q_i \ll S_\tau$. Expanding the square yields a diagonal term $C(q_i^2 - 2S_\tau q_i)$ which is net negative for all $q_i < 2S_\tau$ satisfied by every bid in EU ETS where $q_i \approx S_\tau/70$ after normalisation. The supply constraint is enforced only by accumulated off-diagonal terms, requiring the many-body bound of equation (9) rather than the standard single-violation condition. As problem size grows to $N \sim 10^4$ bids, the required C scales as $O(1/n_{\text{acc}}\bar{q}^2)$, making penalty-encoded QUBO increasingly ineffective.

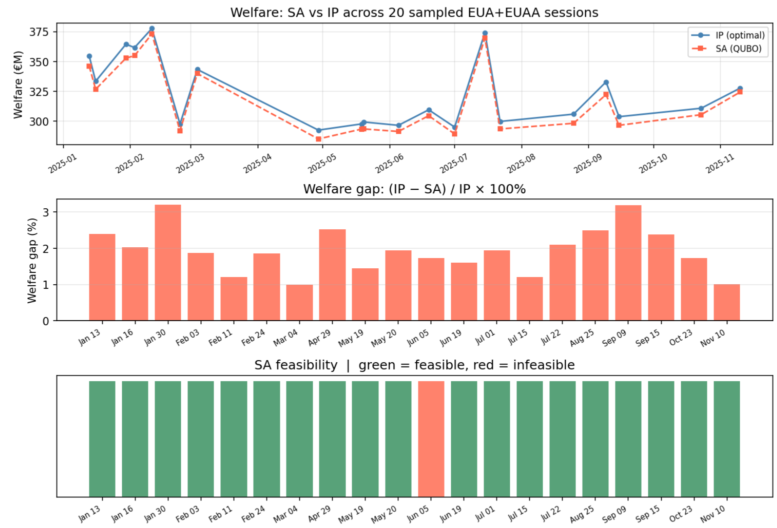

Figure 1: Sample of 20 Random session from the data

Constrained Quadratic Model

A resolution to the penalty scaling problem is D-Wave's Constrained Quadratic Model (CQM) [?], which separates the objective from the constraints and passes both directly to the solver without penalty encoding. The supply constraint $\sum q_i x_i \le S_\tau$ is stated as a native inequality; the solver enforces it exactly without introducing the diagonal imbalance identified above.

Slack variable encoding was investigated as an intermediate approach: converting $\sum q_i x_i \le S_\tau$ to equality via a binary-encoded slack $s = \sum_k s_k 2^k / 2^m$. This fails for the same structural reason the slack bits must span $[0, 1]$ in normalised units, requiring the largest bit to have value $2^{m-1}/2^m \approx 0.5$, which is $35\times$ larger than a typical bid coefficient $\bar{q} \approx 0.014$. The cross terms $2C\, q_i\, s_k$ between bid variables and large slack bits reproduce the original scaling imbalance in a different location. Uniform slack spacing avoids this but requires $1/q_{\min} \approx 10{,}000$ bits per type to account for the smallest bids, making the QUBO intractable.

CQM requires access to `LeapHybridCQMSampler` on D-Wave Leap quantum hardware, a classical–quantum hybrid architecture combining branch-and-bound with quantum annealing for constraint satisfaction. This is the natural path for scaling the current formulation beyond the toy instance.

EUPHEMIA Electricity Market

The EU ETS primary auction is a uniform-price sealed-bid mechanism in which the clearing decision reduces to sorting bids by price and accepting from the top until supply is exhausted an $O(N \log N)$ operation. The combinatorial structure introduced here via package bids is a modelling extension rather than an intrinsic property of the mechanism.

A structurally harder target is the EUPHEMIA algorithm, which clears the pan-European day-ahead electricity market across 25 countries. EUPHEMIA processes block bids all-or-nothing orders spanning contiguous delivery hours whose acceptance is conditional on the clearing prices of all hours in the block simultaneously. This creates mandatory pairwise couplings J_{ik} between block bids in the Ising formulation that cannot be decomposed by period or country. The problem is NP-hard in the general case and the current EUPHEMIA implementation uses heuristics without optimality guarantees. An Ising formulation of block bid clearing is a well-defined and computationally motivated extension of the present work.

Extensions Within Current Framework

Three extensions are identified within the existing QUBO structure. First, a 9-type model treating each fund pool (IF, IX, MF, MX, SF) as a separate credit type τ with its own supply cap S_τ, using the fund allocation columns already present in the EEX dataset. Second, degree-3 package terms for three-way EUA–EUAA–EUA_DE compliance bundles, reducible to QUBO via the Boros–Hammer method [4] at one auxiliary variable per triple, expanding the variable count from N to $O(N\bar{p})$ where $\bar{p}$ is mean bundle size. Third, multi-session calibration using all 213 sessions in the 2025 dataset to test parameter stability across the observed price range of 70–84 per tCO_2.

Full iteration history is available at https://github.com/satkarjuneja/Trinity_Challenge.

References

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